StanfordTriad: a multimodal triadic redesign corpus for privacy-preserving behavioural analysis

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[1] Radboud University · SYNAPSIS. Original corpus: J. A. Edelman, Stanford University (2011).

Abstract

StanfordTriad is a multimodal corpus of small design teams (triads) performing a product-redesign task, created at Stanford University's Center for Design Research (CDR) by Jonathan A. Edelman [1]. Fourteen three-person teams were each recorded performing a single, roughly thirty-minute redesign task engineered to elicit *radical breaks*—moments where a team departs sharply from a provided artifact rather than merely refining it. Each session was captured on four synchronised video streams and transcribed. This descriptor summarises the study design and documents the derived **continuous acoustic features** redistributed here for the summer school. The corpus is gesture-rich, multi-person and speech-bearing, which makes it an ideal stress case for *masking* (de-identification) and for benchmarking pose and gesture pipelines on privacy-preserving data.

1 Background & Summary

Audiovisual recordings of design teams are among the richest behavioural data in the social sciences—and among the hardest to share, because faces and voices identify participants directly. The StanfordTriad corpus was created to study how the *media* teams use (engineering drawings, sketches, rough physical models) shape the *behaviour* that produces incremental versus radical redesign [1]. For the MDIG Multiscale Summer School it serves as the running example across the week: motion tracking, acoustic analysis, pose estimation, and—on the masking day—de-identification with MaskAnyone and utility benchmarking with MaskBench.

2 Methods

2.1 Study design

Fourteen **three-person teams (triads)** each carried out a **single redesign task of $\sim$30 minutes**. Teams were, with one exception, composed of people already familiar with one another, to keep interaction natural. The task was designed to *trigger radical breaks*: teams were presented with an engineering drawing / model of a device (a "material analyzer") and an **open prompt** to move the design forward. The internal study codename was "Envision" (not disclosed to participants). The dissertation analyses four contrasting teams in depth—Group 2 *Optimizers*, Group 7 *Wild Ideas*, Group 9 *User-Centered*, and Group 10 *Breakers*.

2.2 Recording

Sessions were recorded in the CDR **Design Observatory** [2] using **four synchronised video streams**: one camera on *each of the three participants* plus one **top-down camera** over the shared table. Recordings were converted for transcription (`.mov`→`.wav`) and analysed with VCode; time-stamping and direct on-video annotation used Apple Compressor and Apple Motion. Sessions were transcribed with time codes for behavioural coding.

2.3 Behavioural signal

The behaviour of interest is intrinsically **multimodal**: co-speech gesture and enactment, sketching while narrating, joint attention around shared media, and speech. This is precisely why the corpus doubles as a masking challenge—multiple people per frame, frequent occlusion, gesture as the primary signal, and voice as an identifier that needs its own treatment.

3 Data Records

The corpus is accompanied by **derived continuous acoustic features** (not raw audio or video) and the extraction scripts, provided in the repository at `Datasets/StanfordTriad/`. Provenance and the full methodology are in the source dissertation (see *Data Availability*).

Path	**Contents**
`TS_acoustics/`	Continuous acoustic feature time-series, one set per diarised speaker channel (`SPEAKER_00–03`, plus `UNKNOWN`).
`f0_*.csv`	Fundamental frequency (F0 / pitch) over time.
`cog_*.csv`	Spectral centre of gravity (brightness / timbre) over time.
`env_*.csv`	Amplitude envelope (normalised); loudness / intensity over time.
`scripts/continuousacoustics/`	Scripts used to extract the features above.

Speaker channels are the output of diarisation; `UNKNOWN` collects unattributed speech. Features are frame-wise time series suitable for prosodic analysis and alignment with the gesture and pose streams.

4 Technical Validation & Notes

The features here are extracted signals: they carry no faces and no intelligible speech, but the *speaker structure* (who spoke when) is retained. They should therefore be handled with the same care as the underlying recordings. Any reuse of the source video must respect the original participants' consent and the NonCommercial licence below.

5 Usage Notes: masking & benchmarking

On the masking day the corpus is used end-to-end: **MaskAnyone** de-identifies each person (blur, silhouette, skeleton, or face-swap, chosen per subject); **MaskBench** then compares several pose estimators on the masked video to *maximise analytic utility*—selecting the pipeline that preserves the most kinematic signal before any downstream analysis is run. Those privacy-preserving keypoints are exactly what the open **EnvisionBOX** gesture-detection challenge consumes, so the corpus links de-identification directly to shared, reusable science.

6 Data Availability

The source dissertation—treated here as the preprint of record for the corpus—is openly available from Stanford University:

- Edelman, J. A. (2011). *Understanding Radical Breaks: Media and Behavior in Small Teams Engaged in Redesign Scenarios.* PhD dissertation, Dept. of Mechanical Engineering, Stanford University. `http://purl.stanford.edu/ps394dy6131`

References

[1] Edelman, J. A. (2011). *Understanding Radical Breaks: Media and Behavior in Small Teams Engaged in Redesign Scenarios.* PhD dissertation, Stanford University. `http://purl.stanford.edu/ps394dy6131`

[2] Tang, J. C. & Leifer, L. J. (1988). A framework for understanding the workspace activity of design teams. *Proc. CSCW '88*, 244–249.